

Chapter 162 — Radiation Therapy for Prostate Cancer — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

1. **Prostate cancer $\alpha/\beta \approx 1.5$ Gy**, lower than the rectum (~ 3) → large fractions favour tumour kill.
2. **Moderate hypofractionation 60 Gy/20** noninferior (CHHiP, PROFIT); **SBRT 36.25 Gy/5** standard (PACE-B).
3. **Dose escalation**: 6 randomised trials → better biochemical control, **no overall survival (OS) gain**.
4. **ADT (NCCN)**: favourable intermediate none; **unfavourable intermediate 4–6 months; high risk 24–36 months** (12–36 with BT + EBRT).
5. **Phoenix failure = nadir + 2 ng/mL**; ASTRO 1996 = three consecutive rises.
6. **LDR quality**: D90 >90% of prescription; V100 >85%; imaging at day 14–30.
7. **BT relative contraindications**: IPSS >20, Qmax <10 mL/s, TURP <3–6 months, median lobe, pubic arch.
8. **ASCENDE-RT**: LDR boost halves biochemical failure (HR 2.04) but grade 3 GU 18.4% vs 5.2%.
9. **Bounce** >0.2 ng/mL then falls; **monitor to 30 months**; biopsy if still rising after 30 months.
10. **Post-RT biopsy at 30–36 months**; never Gleason-grade marked radiation effect.

BACKGROUND AND TECHNOLOGIC ADVANCES

- **Conventional fractionation**: 1.8–2.0 gray (Gy)/day; megavoltage linear accelerators 6–20 MV; late toxicity = ≥ 90 days after treatment.
- **α/β** : late normal tissue 2–4 Gy (~ 3 , fraction-sensitive); most tumours ≥ 8 (~ 10); **prostate 1.0–1.7**.

CONTEMPORARY TECHNIQUES AND IMAGING

- **MRI misses cancers <10 mm** → systematic $\pm$ targeted biopsy; cores from one target count as one.
- **MRI has no electron density** → must fuse with CT to plan; most useful with hip prosthesis or small gland.
- **PSMA PET/CT** (prostate-specific membrane antigen) before treatment: unfavourable intermediate and high risk.
- **IMRT (intensity-modulated radiation therapy)**: commonest curative EBRT; no randomised trial vs 3D-CRT (three-dimensional conformal radiotherapy).
- **GTV** (gross tumour volume) = dominant lesion on MRI; **CTV** (clinical target volume) = + whole prostate $\pm$ seminal vesicles $\pm$ nodes; **PTV** (planning target volume) = + set-up/motion margin.
- **Margins**: IGRT (image-guided RT) 3–7 mm; SBRT (stereotactic body RT) 3–5 mm; MRI-guided 2 mm vs CT-guided 4 mm (MIRAGE).
- **Moderate hypofractionation** 2.25–4 Gy/fraction; **ultrahypofractionation** ≥ 5 Gy (5–10 Gy).
- **HYPO-RT-PC**: 42.7 Gy/7 = 78 Gy/39 (noninferior; late toxicity equal).
- **SBRT ceiling**: no gain above 40 Gy/5 (36 Gy/4); more late toxicity.
- **SBRT**: every other day; commonest U.S. schedule 725 cGy $\times$ 5; CyberKnife needs fiducials.
- **FLAME**: boost to dominant lesion → better biochemical progression-free survival (HR 0.45), no added toxicity.
- **Protons**: Bragg peak; spread-out Bragg peak (SOBP) for tumour thickness; ↓ low-dose bath, **not** high-dose volume.
- **Proton trap: femoral head dose higher** (lateral beams) → avascular necrosis risk.
- **PROG 95-09**: 79.2 vs 70.2 Gy-equivalent → fewer biochemical failures; no photon comparator.
- **No randomised trial of protons vs IMRT**; PARTIQoL (EPIC bowel score at 24 months) ongoing.

BRACHYTHERAPY

- **Isotope half-lives**: ^{125}I 59.6 d; ^{103}Pd 17 d; ^{131}Cs 9.7 d.
- **Monotherapy dose**: ^{125}I 140–160; ^{103}Pd 110–125; ^{131}Cs 100–115 Gy.

- **Combined**: EBRT 45 Gy + implant at 60–70% of monotherapy dose.
- **LDR**: energy ~ 21 –30 keV. **Peripheral loading** spares the urethra. **Intraoperative planning** gives best coverage and lowest urethral dose.
- **Rectal RV100**: <1 cm³ ($^{125}\text{I}/^{103}\text{Pd}$), <0.5 cm³ (^{131}Cs); <1.0 cm³ on day 1 / <1.3 cm³ on day 30.
- **HDR (high dose rate)**: ^{192}Ir , 380 keV, remote afterloader; no post-implant dosimetry; needs a shielded room.
- **HDR monotherapy**: 27–28 Gy/2; 31.5 Gy/3; 34–38 Gy/4.
- **Single 19 Gy HDR fails**: 5-year local failure 29% vs 3% with 27 Gy/2.
- **Prerequisites**: fit for general anaesthesia (standard) and TRUS.
- **Cytoreduction**: ADT up to $\sim 30\%$; 5 α -reductase inhibitor $\sim 20\%$ (less morbid).
- **RTOG 0232**: intermediate risk, BT alone = EBRT + BT for control; less toxicity → **BT alone standard**.
- **UK HDR boost trial**: EBRT $\pm$ HDR 17 Gy/2 → 21% absolute relapse-free survival gain at 12 years.
- **Gleason 9–10**: EBRT + BT lowest 5-year prostate cancer mortality (3% vs 12% RP vs 13% EBRT).

METASTATIC DISEASE

- **Oligometastatic** ≤ 5 lesions; **oligoprogression** → treat sites, keep systemic therapy.
- **CHAARTED high volume**: ≥ 4 bone metastases with ≥ 1 outside spine or pelvis, or visceral.
- **STAMPEDE (metastatic)**: prostate RT improves OS only in **low burden** (HR 0.68).
- **HORRAD**: prostate RT, no OS benefit overall.
- **^{223}Ra (ALSYMPCA)**: alpha; 50 kBq/kg every 4 weeks $\times$ 6; OS +2.6 months; no visceral metastases (nodes <3 cm).
- **^{177}Lu -PSMA-617 (VISION)**: post-taxane PSMA+ castration-resistant disease; OS 15.3 vs 11.3 months (HR 0.62).
- **^{89}Sr** : pain relief, rarely pain-free; **myelosuppression**.
- **SABR-COMET**: 1–5 metastases; 5-year OS 42.3% vs 17.7%.
- **ORIOLE**: 1–3 metastases; progression at 6 months 19% vs 61%.
- **STOMP**: ADT-free survival 21 vs 13 months — **not significant**.
- **ESTRO-ACROP consensus**: metastasis-directed therapy for ≤ 5 lesions; oligoprogressive castration-resistant disease → **do not change systemic therapy**.

ANDROGEN SUPPRESSION WITH RADIATION

- **Mechanism**: castration ↓ Ku70 → impaired repair; RT ↑ androgen receptor signalling.
- **TROG 96.01**: 6 months (not 3) ADT → better OS.

- **RTOG 94-08:** + 4 months ADT benefits **intermediate, not low** risk.
- **DFCI 95-096:** + 6 months ADT → OS benefit only with no or minimal comorbidity.
- **EORTC 22863:** 3 years goserelin + 70 Gy → better OS.
- **RTOG 86-10:** 4 months ADT → better disease-specific mortality, **not OS**.
- **SPCG-7 / PR.3:** adding RT to ADT → OS +8% (7 years) to +10% (10 years).
- **RTOG 92-02:** long ADT → OS benefit only in **Gleason 8–10**.
- **PCS IV:** 18 months = 36 months; better quality of life with 18.
- **RTOG 99-10:** mostly intermediate risk; longer neoadjuvant ADT no benefit.
- **RADAR:** extra ADT benefit falls as dose rises; none with HDR boost.

PELVIC NODAL TREATMENT

- **RTOG 94-13 / GETUG-01:** intact prostate → no durable benefit.
- **SPPORT (RTOG 0534):** postoperative pelvic RT + short ADT → fewer distant metastases.
- **Superior border L4/L5 (RTOG 09-24):** old L5/S1 border misses common iliacs.

EVALUATING THE RESPONSE

- **Median nadir:** ~0.6 ng/mL (70 Gy); ~0.3 (79 Gy); ≤0.1 (EBRT + BT).
- **No evidence of disease (NED) nadir 0.4–0.5;** local failure >1.0; distant failure >2.
- **Biochemical failure <18 months** → distant metastasis and prostate cancer death.
- **Nadir >2** at 1 year, >1.5 at 2 years, >0.5 → worse outcome.
- **After BT:** nadir <0.5; PSA ≤0.2 at 4–5 years → durable cure.
- **Doubling time:** distant 3–6 months; local ≥1 year; **<3 months** → prostate cancer mortality; <12 months → consider intervention.
- **Bounce:** ~½ after LDR (⅓ if young); lasts 12–18 months; **approaching 10 ng/mL** → systemic staging.
- **Post-RT recurrence imaging (ACR):** bone scan first; CT/MRI if negative. AUA: mpMRI after a negative biopsy.
- **High-molecular-weight cytokeratin** = basal cells → benign atypia, not cancer.
- **Box 162.1 score:** 0–2 minimal; 3–4 moderate; 5–6 marked (marked = behaves as negative).

TREATMENT EFFICACY

- **RTOG 0126:** 79.2 vs 70.2 Gy; biochemical failure-free survival 80% vs 65%; OS HR 1.00.
- **MD Anderson:** 78 vs 70 Gy; freedom from failure 73% vs 50%.
- **CHHiP:** 60/20 noninferior; **57/19 not**.
- **RTOG 04-15 (low risk):** 70 Gy/28 noninferior; more late toxicity.
- **HYPRO:** 64.6 Gy/19 **not superior**.
- **Postoperative:** 62.5 Gy/25 noninferior toxicity to 66.6 Gy/37.

- **SBRT rectal risk:** >35% of rectal circumference receiving >39 Gy.

BT OUTCOMES AND COMPARISON

- **No randomised trial of BT alone vs EBRT alone;** retrospective data favour BT biochemically; **EBRT <72 Gy inferior**.

MORBIDITY

- **IMRT vs 3D-CRT:** ↓ GI, **not** GU toxicity.
- **Rectal V70 >25%** → more rectal complications.
- **Urinary symptoms** peak at 2 months; baseline by 12 months; incontinence uncommon.
- **Pre-BT IPSS >20** → ~29% retention; <10 → ~2%.
- **Proctitis** at 6–18 months: steroid suppositories or sucralfate → argon plasma coagulation or hyperbaric oxygen → colostomy.
- **BT rectal complications** 1–2%; colostomy ~0.3%.
- **Spacer (Denonvilliers' fascia):** late rectal toxicity 2% vs 7% (hydrogel); 2.9% vs 13.8% (hyaluronic acid).
- **ProtecT:** RT → more bowel bother, bloody stools; **no faecal incontinence difference**.
- **Potency best with BT alone** (~76% at 5 years); ADT worsens it.
- **PDE5 inhibitors** effective in ~¾.

SALVAGE

- **Prerequisite:** biopsy-proven, local-only failure.
- **Salvage prostatectomy (CALGB 9687):** ~39% salvaged at 5 years; ~63% incontinent.
- **BT favoured** for salvage; NRG 0526: 68% salvaged at 5 years.
- **HDR salvage:** 8 Gy × 4; strictures a key toxicity.

CLASSIC TRAPS

- **Trap:** dose escalation improves survival — **Actually:** biochemical control only.
- **Trap:** LDR means low dose — **Actually:** low dose *rate*; permanent seeds.
- **Trap:** protons spare every organ — **Actually:** femoral head dose higher; high-dose rectum unchanged.
- **Trap:** a PSA rise after BT means failure — **Actually:** probable bounce; monitor to 30 months.
- **Trap:** ASTRO "three rises" is current — **Actually:** Phoenix nadir + 2 ng/mL.
- **Trap:** Gleason-grade the irradiated biopsy — **Actually:** only if radiation effect minimal.
- **Trap:** single-fraction HDR 19 Gy suffices — **Actually:** 29% local failure; multifraction standard.
- **Trap:** IMRT lowers GU toxicity — **Actually:** GI only; urethra inside target.
- **Trap:** pelvic nodal RT proven in intact prostate — **Actually:** RTOG 94-13 and GETUG-01 negative.
- **Trap:** biopsy at 6–12 months post-RT — **Actually:** postmitotic death; wait 30–36 months.